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CASE STUDY REPORT

Case Study 3: Logistic Regression on the Titanic Dataset
Case Study 6: Bank Marketing Classification

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CASE STUDY 3

Logistic Regression on the Titanic Dataset

1. Problem Statement

The sinking of the Titanic in 1912 is one of the most well-known disasters in history. Out of the passengers on board, only some of them survived. This case study tries to find out which passengers were more likely to survive and which were less likely. For this, a Logistic Regression model is built that uses passenger details such as class, gender, age, family members on board, ticket fare and port of boarding to predict whether a passenger survived or not.

The study also explains how each of these details changes the chance of survival, so that the result is easy to understand and not just a prediction.

2. Dataset Description

The dataset has records of 891 passengers and 12 columns. The target column is **Survived**, where 1 means the passenger survived and 0 means the passenger did not survive. The other columns are described in Table 1.

Table 1: Columns of the Titanic dataset

Column	**Meaning**
PassengerId	Serial number of the passenger
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Ticket class (1st, 2nd or 3rd)
Name, Ticket	Passenger name and ticket number
Sex	Gender of the passenger
Age	Age in years
SibSp	Number of siblings or spouses on board
Parch	Number of parents or children on board
Fare	Price paid for the ticket
Cabin	Cabin number
Embarked	Port of boarding (C, Q or S)

When the data was checked, three columns had missing values: **Age** had 177 missing values, **Cabin** had 687 missing values and **Embarked** had 2 missing values. Out of 891

passengers, 342 survived (38.38%) and 549 did not survive (61.62%).

3. Methodology

The work was carried out in Python using the libraries pandas and NumPy for handling data, Matplotlib for graphs and scikit-learn for building the model. The steps followed are given below.

- **Loading the data:** The Excel file was read into a pandas table and its size, column types, summary values and missing values were checked.
- **Exploring the data:** Graphs and correlation values were used to see how survival is related to gender, class, age and fare.
- **Handling missing values:** The missing ages were filled with the median age, and the two missing ports were filled with the most common port. The Cabin column was dropped because most of its values were missing.
- **Removing unnecessary columns:** PassengerId, Name and Ticket were removed because they do not help in predicting survival.
- **Converting text to numbers:** Sex and Embarked were converted into 0/1 columns so that the model can use them.
- **Splitting the data:** 80% of the passengers (712) were used for training and 20% (179) for testing. The split keeps the same survival ratio in both parts.
- **Building the model:** A Logistic Regression model was trained on the training part.
- **Evaluating the model:** The model was tested on the unseen 179 passengers using accuracy, precision, recall, F1-score, confusion matrix and ROC curve.

4. Statistical Findings

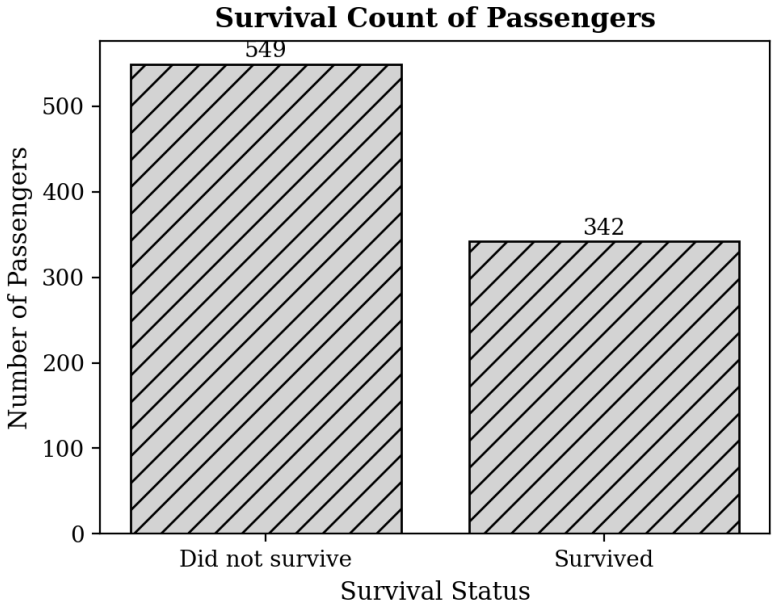


Figure 1: Survival count of passengers

Interpretation: The bar graph shows that more passengers died than survived. Only 342 passengers (38.38%) survived, while 549 (61.62%) did not. The classes are not perfectly balanced, but the difference is small enough for the model to learn from both groups.

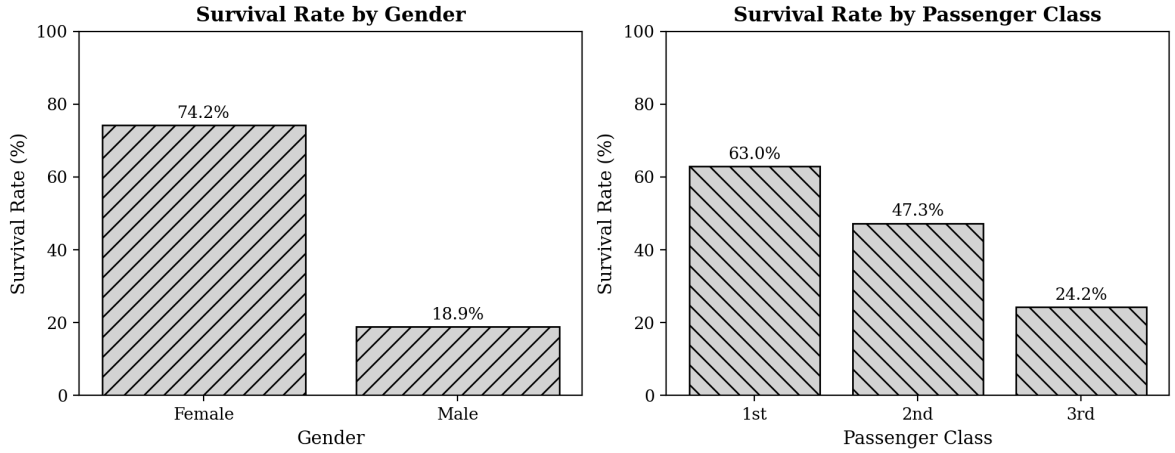


Figure 2: Survival rate by gender and by passenger class

Interpretation: Gender and class made a very large difference. About 74.20% of the women survived, but only 18.89% of the men did. In the same way, 62.96% of first class passengers survived, compared to 47.28% in second class and only 24.24% in third class. This agrees with the "women and children first" rule followed during the rescue, and it also shows that passengers travelling in higher classes had a better chance.

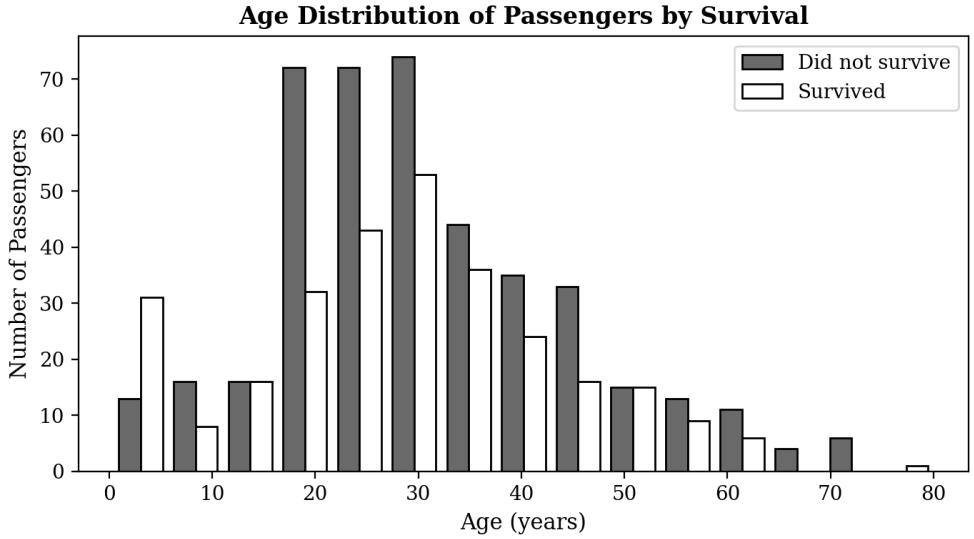


Figure 3: Age distribution of passengers by survival

Interpretation: Most passengers were between 20 and 40 years old. Young children had a better chance of survival; among the 62 passengers younger than 10 years, about 61.3% survived. For passengers aged 60 and above, the survival rate was only about 26.9%. The median age was 28 for both groups, so age alone is a weak factor compared to gender and class.

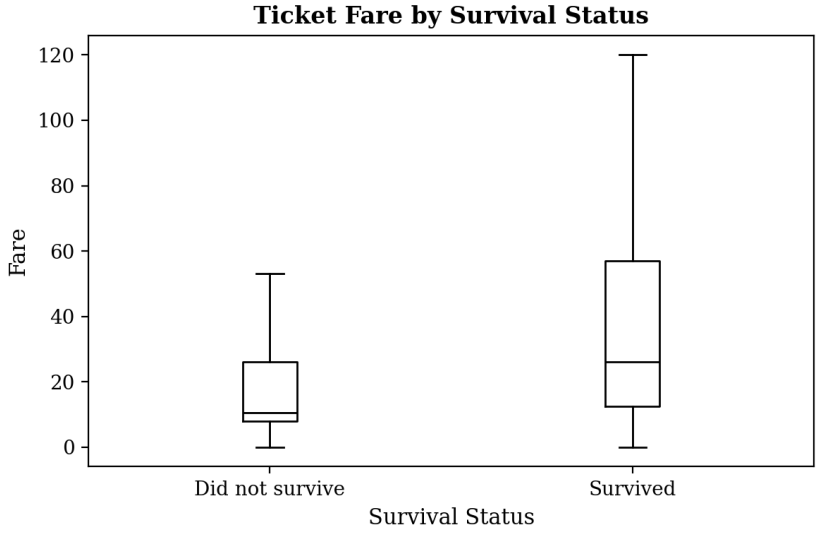


Figure 4: Ticket fare by survival status

Interpretation: Passengers who survived paid higher fares. The median fare of survivors was 26.0, while for those who did not survive it was 10.5. Since higher fares mean better classes and cabins closer to the lifeboats, this agrees with the class result above. Extreme fares were hidden in the plot so that the main boxes can be seen clearly.

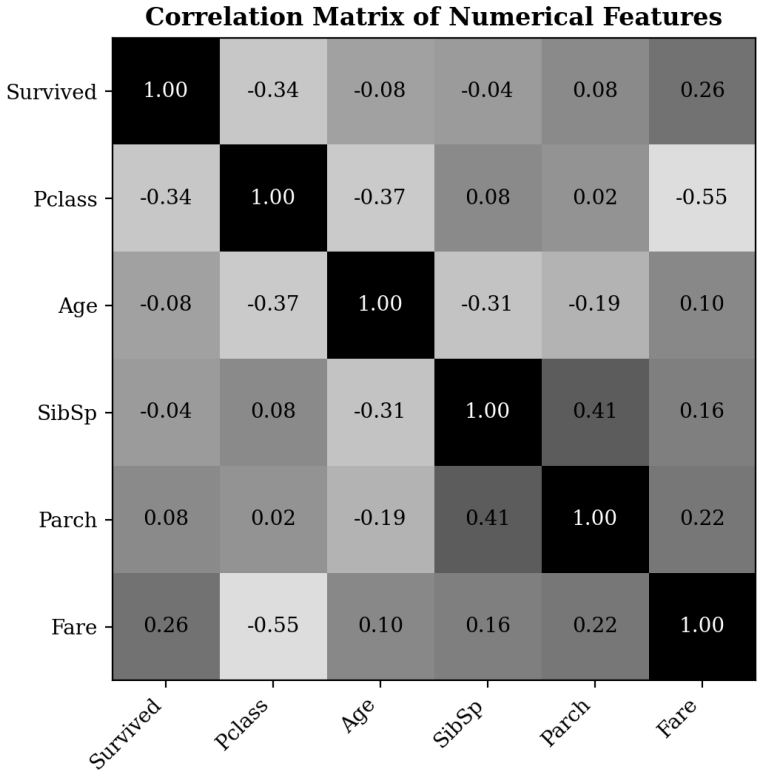


Figure 5: Correlation matrix of numerical features

Interpretation: Survived has a negative correlation of -0.34 with Pclass, which means that as the class number goes up (from 1st to 3rd), the chance of survival goes down. It has a positive correlation of 0.26 with Fare. Pclass and Fare are strongly related to each other (-0.55), because first class tickets cost more. The other features have only weak correlation with survival.

5. Model Evaluation

Table 2: Performance of the Logistic Regression model on test data

Metric	**Value**
Accuracy	0.8045
Precision	0.7931
Recall	0.6667
F1-Score	0.7244
ROC-AUC	0.8443

The model correctly predicted about 80.45% of the passengers in the test set. Its training accuracy was 80.76%, which is almost the same, so the model is not overfitting and is able to work well on new data. Precision of 0.79 means that when the model says a passenger

survived, it is right about 79% of the time. Recall of 0.67 means that the model found about 67% of the passengers who really survived. The F1-score of 0.72 balances these two values.

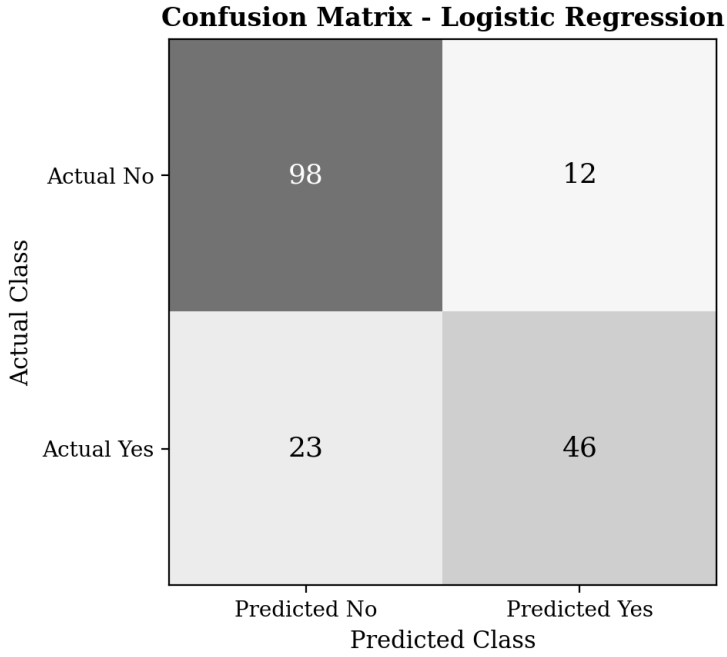


Figure 6: Confusion matrix of the Logistic Regression model

Interpretation: Out of 179 test passengers, 98 non-survivors and 46 survivors were predicted correctly. The model made 12 wrong predictions where it said "survived" for a passenger who did not, and 23 wrong predictions where it missed a passenger who actually survived. The model is therefore a little more careful in predicting survival, which is why recall is lower than precision.

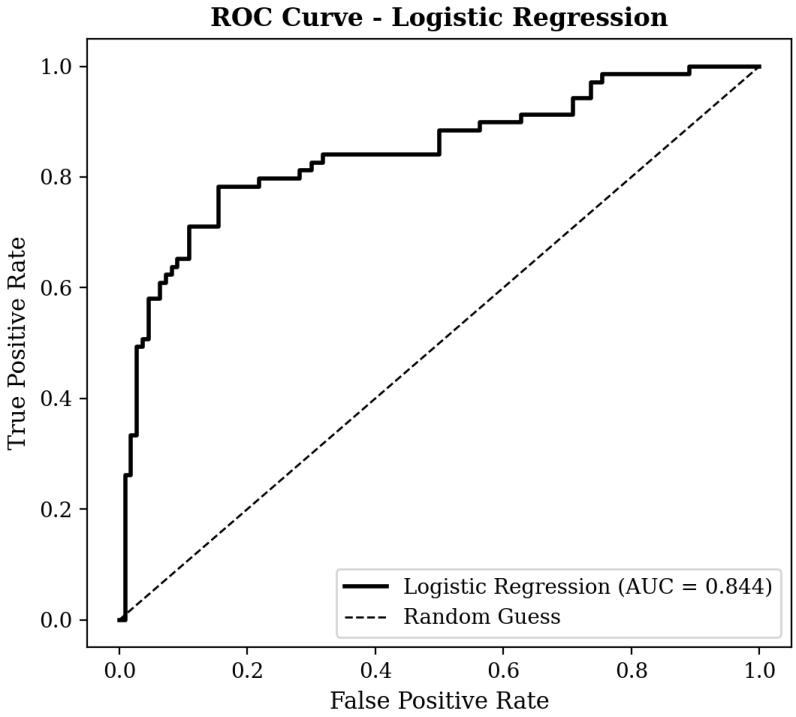


Figure 7: ROC curve of the Logistic Regression model

Interpretation: The ROC curve stays well above the dashed line, which stands for random guessing. The area under the curve (AUC) is 0.844. This means that if one survivor and one non-survivor are picked at random, the model gives a higher survival chance to the survivor about 84% of the time, which is a good result.

6. Interpretation of the Model

Table 3: Coefficients and odds ratios of the model

Feature	Coefficient	Odds Ratio
Embarked_Q	0.2812	1.3247
Fare	0.0022	1.0022
Age	-0.0385	0.9622
Parch	-0.0709	0.9316
SibSp	-0.2438	0.7836
Embarked_S	-0.3828	0.6819
Pclass	-1.0919	0.3356
Sex_male	-2.5579	0.0775

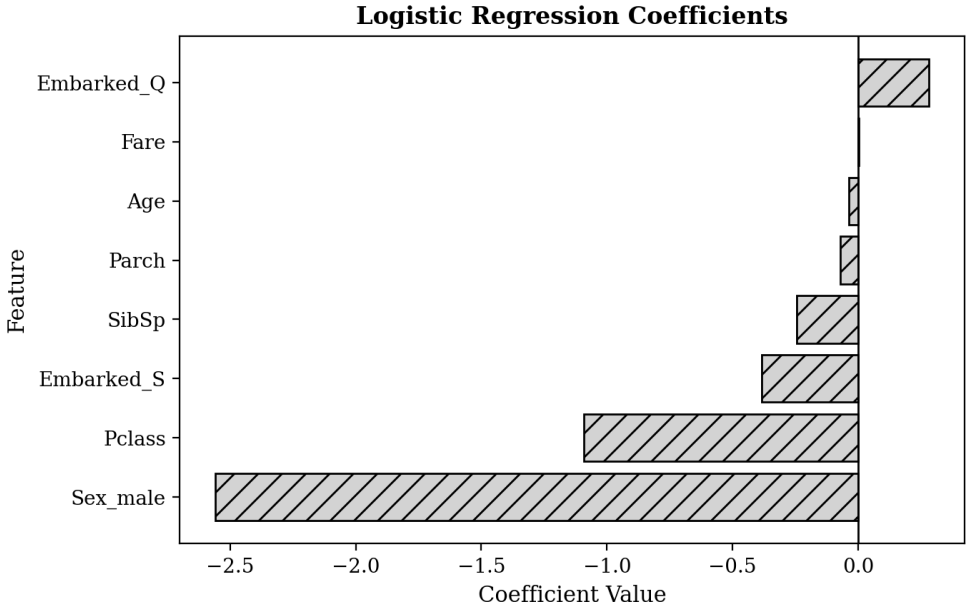


Figure 8: Coefficients of the Logistic Regression model

Interpretation: A negative coefficient lowers the chance of survival and a positive one increases it. The odds ratio shows how many times the odds change when the feature increases by one unit.

- **Sex_male (-2.56, odds ratio 0.08):** This is the strongest factor. The odds of survival for a man were only about 8% of the odds for a woman, keeping other things the same.
- **Pclass (-1.09, odds ratio 0.34):** For each step down in class (for example from 2nd to 3rd), the odds of survival fall to about one third.
- **SibSp (-0.24, odds ratio 0.78):** Passengers travelling with more siblings or a spouse had slightly lower odds of survival.
- **Age (-0.04, odds ratio 0.96):** Every extra year of age reduces the odds of survival by about 4%, so younger passengers did better.
- **Embarked_S (-0.38) and Embarked_Q (+0.28):** Compared to passengers who boarded at Cherbourg, those who boarded at Southampton had lower odds and those from Queenstown had slightly higher odds.
- **Fare and Parch:** Their effect is very small once the other features are included in the model.

The trained model was also saved and a small Streamlit application (app.py) was prepared, in which the user can enter passenger details and get the survival probability.

7. Conclusion

The Logistic Regression model predicted Titanic survival with an accuracy of 80.45% and an ROC-AUC of 0.844. The results show that gender was the most important factor, followed by passenger class, and that younger passengers and those with fewer family members on board had a slightly better chance. The model is simple, easy to explain and gives reliable results on new data.

CASE STUDY 6

Bank Marketing Classification

1. Problem Statement

A Portuguese bank ran phone-call marketing campaigns to sell a term deposit to its customers. Calling every customer takes a lot of time and money, and most customers say no. The aim of this case study is to build classification models that can predict whether a customer will subscribe to a term deposit, so that the bank can contact the right customers and save effort.

Four models are built and compared: Logistic Regression, Decision Tree, K-Nearest Neighbours (KNN) and Random Forest. The study then explains which customer details matter the most and gives simple business suggestions.

2. Dataset Description

The dataset (Moro et al., 2011) contains 45,211 customer records and 16 input columns, along with the target column **y**, which tells whether the customer subscribed (yes) or not (no). Table 4 lists the input columns.

Table 4: Input columns of the bank dataset

Group	Columns
Customer details	age, job, marital, education, default, balance, housing, loan
Last contact of this campaign	contact, day, month, duration
Other details	campaign, pdays, previous, poutcome

There are no empty (null) values and no duplicate rows. However, some columns use the word "unknown" for missing information: job (288 rows, 0.64%), education (1,857 rows, 4.11%), contact (13,020 rows, 28.80%) and poutcome (36,959 rows, 81.75%). These were treated as a separate category instead of being deleted, because removing them would have thrown away a large part of the data.

The target is imbalanced: 39,922 customers (88.3%) said no and only 5,289 customers (11.7%) said yes.

3. Methodology

- **Loading and checking the data:** The file uses semicolons to separate columns. Its size, types, summary values, unknown entries and duplicates were checked.

- **Removing the duration column:** Call duration is known only after the call is finished, so it cannot be used to decide whom to call. Keeping it would make the results look better than they really are, so it was removed.
- **Encoding:** The yes/no columns (default, housing, loan and the target) were changed to 1/0. Other text columns (job, marital, education, contact, month, poutcome) were changed into separate 0/1 columns. This gave 41 input features.
- **Splitting the data:** 80% (36,168 rows) was used for training and 20% (9,043 rows) for testing. The yes/no ratio is the same in both parts (11.7%).
- **Scaling:** The numerical columns (age, balance, day, campaign, pdays and previous) were scaled so that they have similar ranges. The scaler learned only from the training data, so the test data stays unseen. This is especially important for KNN and Logistic Regression.
- **Building the models:** Logistic Regression, a Decision Tree (maximum depth 6), KNN (5 neighbours) and a Random Forest (200 trees, maximum depth 12) were trained. The depth limits stop the tree models from memorising the training data.
- **Comparing the models:** All models were tested on the same test data using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix and ROC curve.

4. Statistical Findings

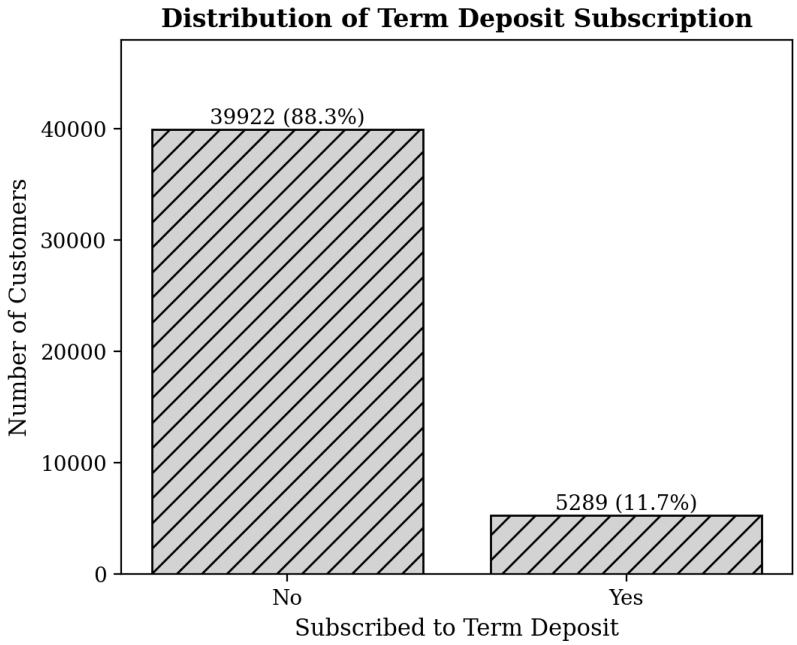


Figure 9: Distribution of term deposit subscription

Interpretation: Only about 1 in 9 customers subscribed. Because of this imbalance, a model that simply says "no" for everyone would already be right 88.3% of the time. So accuracy alone is not a fair way to judge the models, and precision, recall and F1-score for the "yes" group must also be looked at.

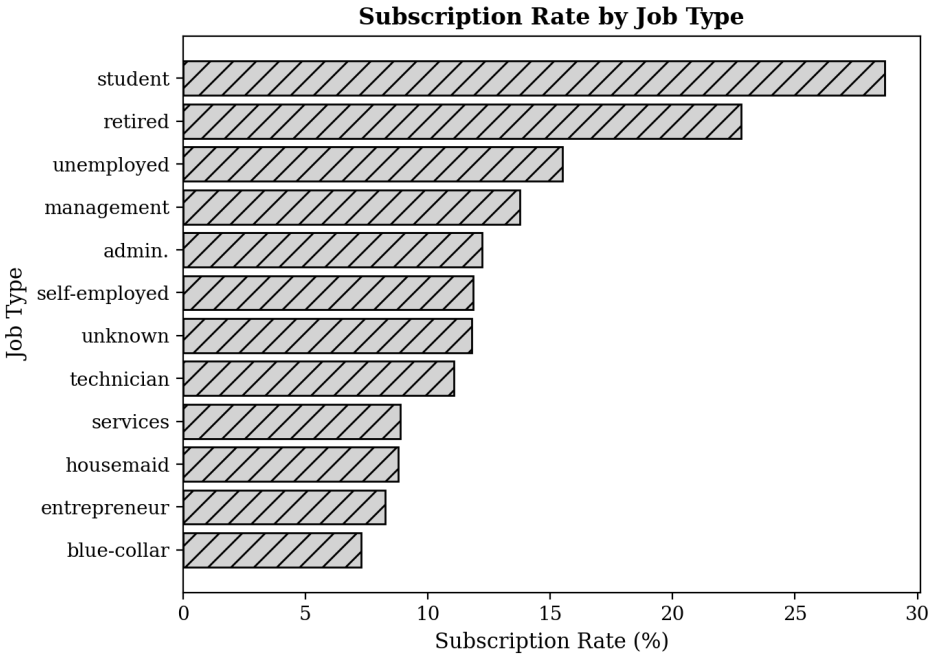


Figure 10: Subscription rate by job type

Interpretation: Students (28.68%) and retired people (22.79%) are the most likely to subscribe, followed by unemployed customers (15.50%). Blue-collar workers (7.27%), entrepreneurs (8.27%) and housemaids (8.79%) are the least likely. Students and retired customers may have savings and fewer financial commitments, so a term deposit suits them.

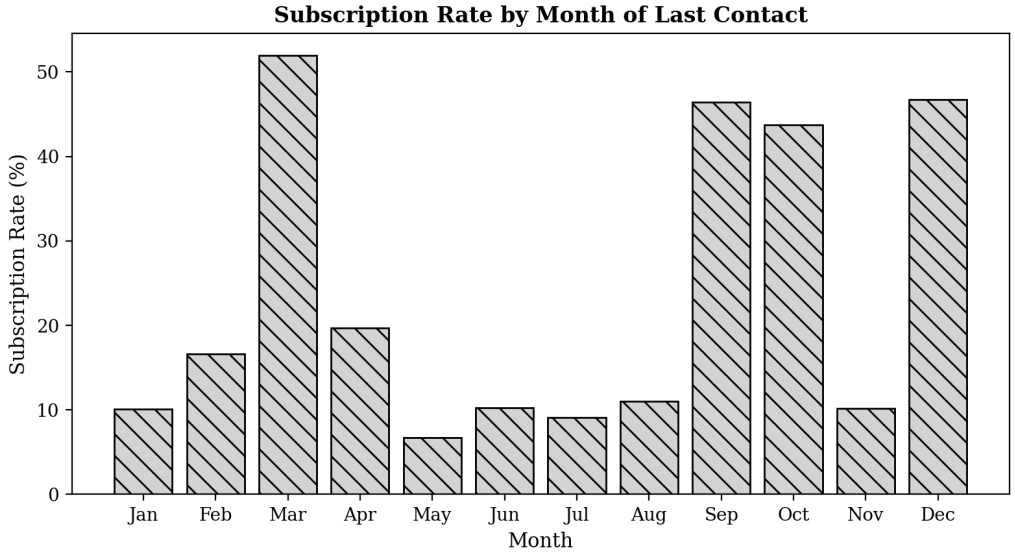


Figure 11: Subscription rate by month of last contact

Interpretation: March (51.99%), September (46.46%), October (43.77%) and December (46.73%) have very high success rates, but these months had very few calls (477, 579, 738 and 214). May had the lowest success rate (6.72%) even though it had the most calls (13,766). So the bank made most of its calls in months where customers were least interested.

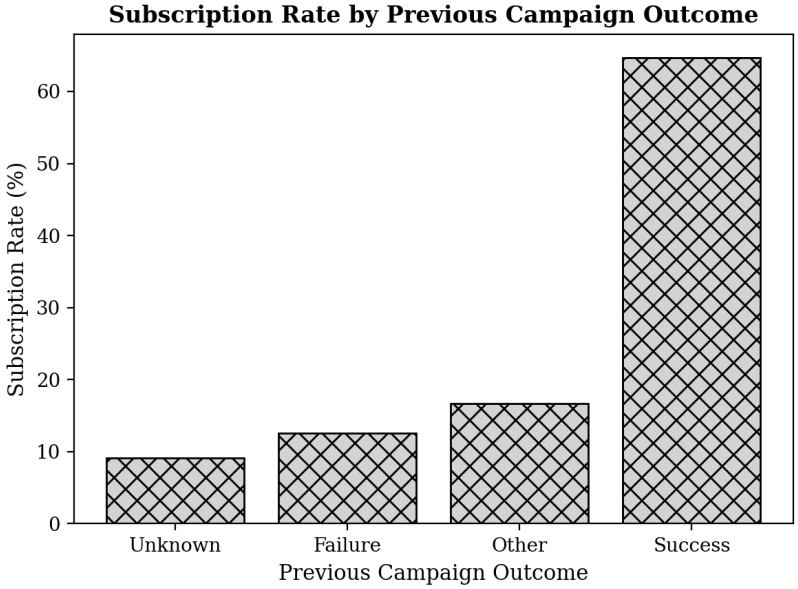


Figure 12: Subscription rate by previous campaign outcome

Interpretation: Customers who had said yes in an earlier campaign subscribed again 64.73% of the time. For customers whose earlier result was unknown, the rate was only 9.16%. This shows that past success is a very strong sign of future success.



Figure 13: Age, balance and number of contacts compared with subscription

Interpretation: The median age is almost the same for both groups (39 for No and 38 for Yes), so age alone does not separate them. Customers who subscribed had a higher median balance (733 euros) than those who did not (417 euros). Subscribers were also contacted fewer times on average (2.1 times compared to 2.8 times), which suggests that calling the same customer again and again does not help.

5. Model Evaluation

Table 5: Performance of the four models on test data

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8933	0.6655	0.1767	0.2793	0.7717
Decision Tree	0.8925	0.6396	0.1862	0.2884	0.7122
KNN	0.8873	0.5520	0.1957	0.2889	0.7045
Random Forest	0.8946	0.6937	0.1777	0.2829	0.7978

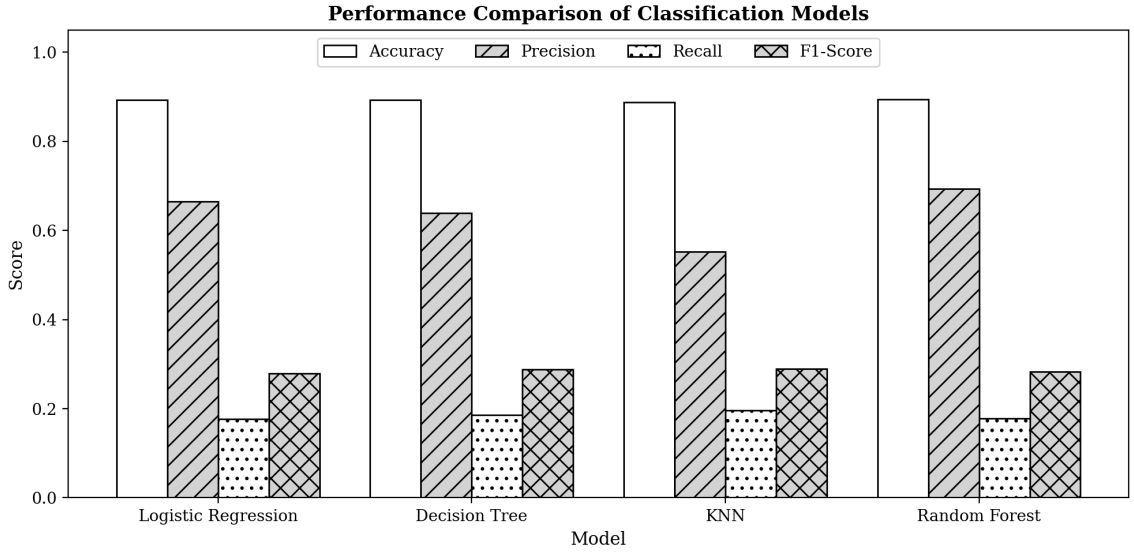


Figure 14: Performance comparison of the four classification models

Interpretation: All four models have a similar accuracy of about 89%. Random Forest is the highest (89.46%), followed by Logistic Regression (89.33%) and Decision Tree (89.25%), while KNN is the lowest (88.73%). Precision is good for Random Forest (0.69), which means that when it predicts a "yes", it is right about 69% of the time. However, recall is low for all models (0.18 to 0.20), which means that they find only about one in five of the customers who really subscribe. This is mainly because of the imbalanced data. KNN has the highest recall (0.1957) but the lowest precision (0.55).

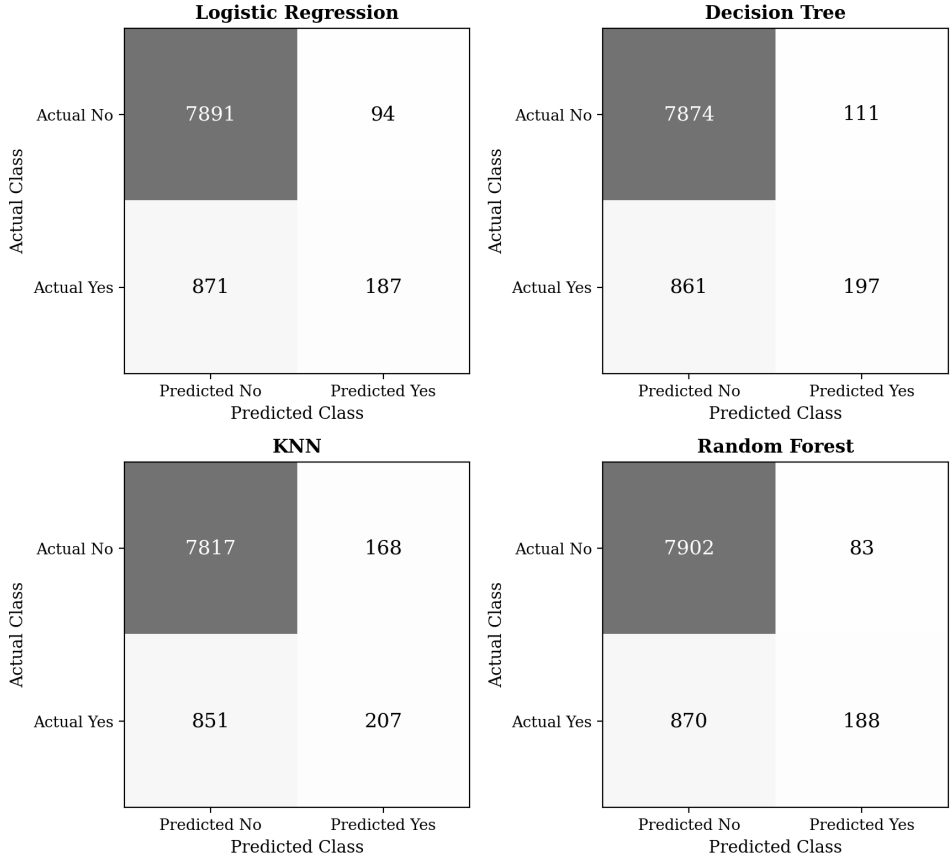


Figure 15: Confusion matrices of the four models

Interpretation: Each model correctly identifies almost all of the 7,985 customers who said no (7,817 to 7,902). But out of 1,058 customers who said yes, they find only 187 to 207. Random Forest makes the fewest false alarms (83 customers wrongly predicted as yes), while KNN makes the most (168). The models are therefore very safe in avoiding useless calls but they miss many real subscribers.

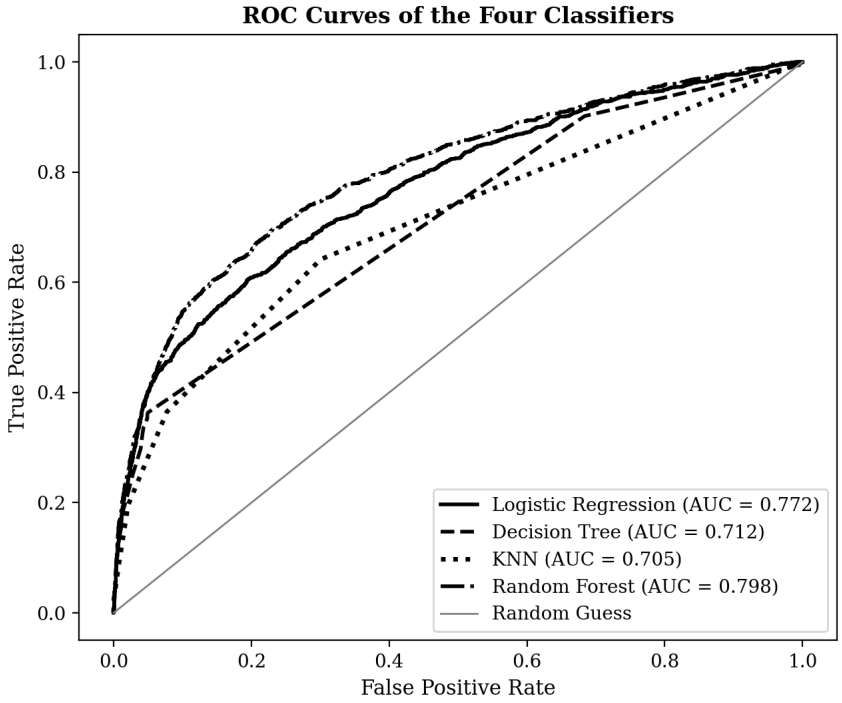


Figure 16: ROC curves of the four classifiers

Interpretation: The ROC curve shows how well a model separates the two groups at every cut-off value. Random Forest has the best AUC (0.798), followed by Logistic Regression (0.772), Decision Tree (0.712) and KNN (0.705). A higher AUC means better ranking of customers, and Random Forest is clearly ahead. So even though the four models look alike in accuracy, Random Forest is the best at telling which customers are more likely to subscribe. It was therefore selected as the best model.

6. Business Recommendations

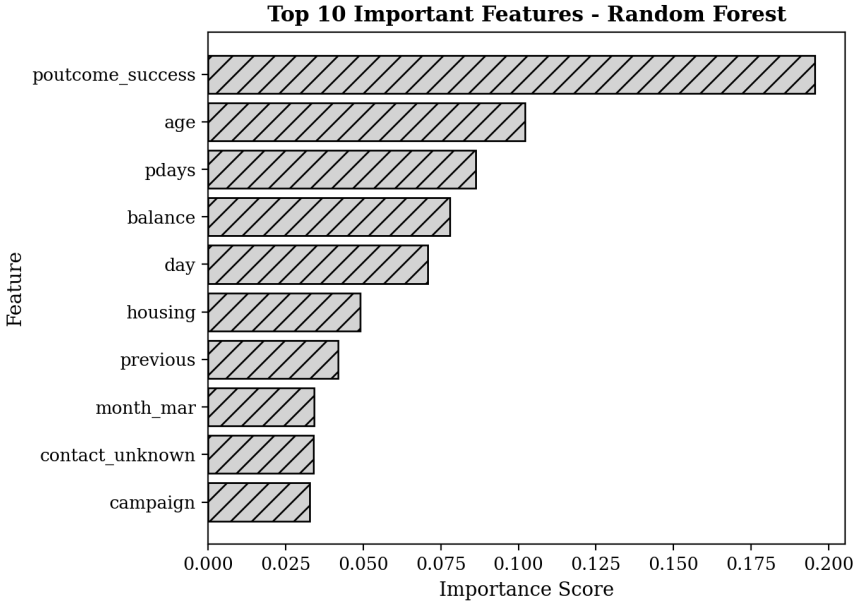


Figure 17: Top 10 important features of the Random Forest model

Interpretation: The most important feature is a successful outcome of the previous campaign (poutcome_success, importance 0.196), which is nearly double that of any other feature. After this come age, pdays (days since the last contact), balance and the day of the month. This agrees with the findings in the earlier graphs.

The model can also rank customers by their chance of subscribing. Table 6 shows what happens if the bank calls only the highest-ranked customers in the test data.

Table 6: Result of calling only the top-ranked customers

Customers Contacted (%)	Subscribers Found	Share of All Subscribers (%)	Subscription Rate (%)
10	449	42.44	49.67
20	639	60.40	35.34
30	748	70.70	27.58
50	881	83.27	19.49

By calling only the top 20% of customers, the bank would reach 60.4% of all subscribers, and about 35 out of every 100 calls would be successful, compared to 11.7% when customers are called without any selection. Based on all the findings, the following suggestions can be given to the bank:

- Call customers who had subscribed in an earlier campaign first, because their success rate is the highest (64.73%).

- Give extra attention to students and retired customers, and to customers with a higher balance and without a housing loan (subscription rate 16.7% compared to 7.7% for those with a housing loan).
- Run more campaigns in March, September, October and December, and reduce the heavy calling in May, when the success rate is only 6.72%.
- Avoid calling the same customer many times, because customers who subscribed needed fewer calls.
- Use the model score to prepare a call list in order of priority, so that the calling team spends its time on the customers most likely to say yes.

7. Conclusion

Four models were compared for predicting term deposit subscription. All of them reached an accuracy of around 89%, but because only 11.7% of the customers subscribe, accuracy is not enough to judge them. Random Forest gave the best ROC-AUC (0.798), the highest precision (0.69) and the highest accuracy (89.46%), and it was chosen as the best model. All the models have a low recall (about 0.18 to 0.20), so they miss many real subscribers when a normal 0.5 cut-off is used. Even so, ranking customers with the model gives a big improvement over random calling. This can be improved further in future by lowering the cut-off value or by balancing the classes.

OVERALL CONCLUSION

In Case Study 3, Logistic Regression explained Titanic survival well (accuracy 80.45%, AUC 0.844) and showed that gender and class were the main reasons behind who survived. In Case Study 6, four classifiers were compared on the bank marketing data and Random Forest turned out to be the best model (AUC 0.798). Both case studies show that good results depend not only on the model but also on proper data cleaning, careful choice of evaluation measures and clear explanation of the results.

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